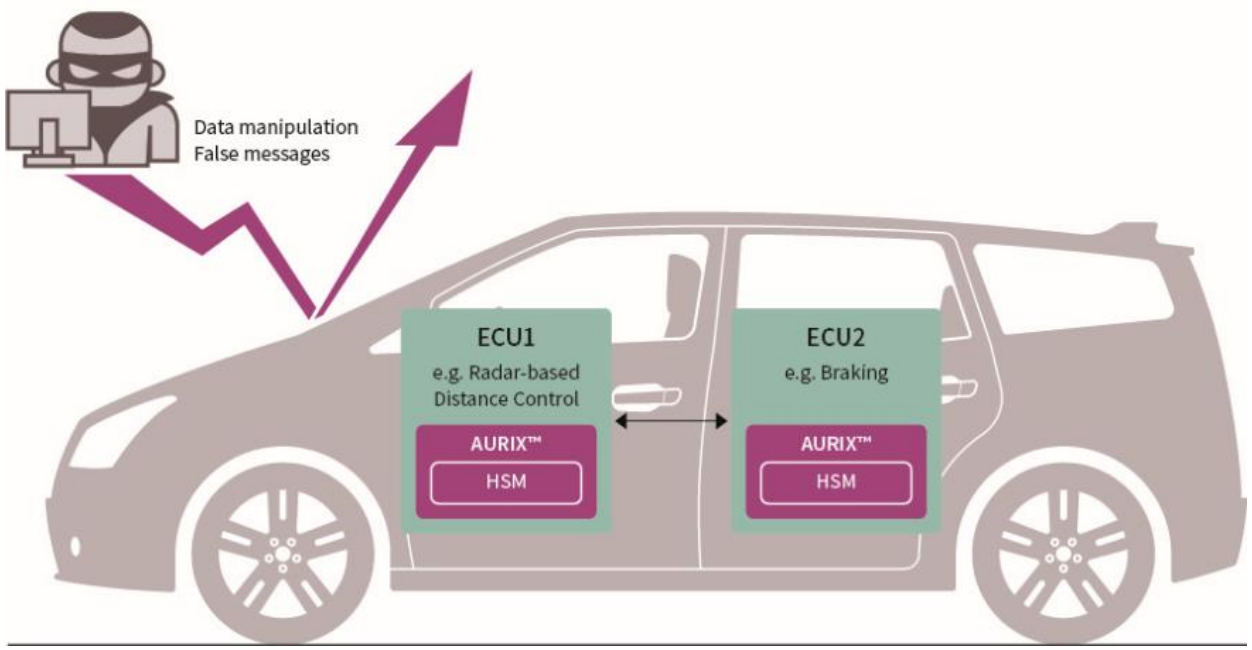
AUTOSAR SecOC [5]

Why Secure Onboard Communication (SecOC)?

SecOC addresses threats, such as,

- Hardware/software manipulation: A hardware/software component is inserted/manipulated to become part of the communication network.

- Control over an ECU: To take control over of an ECU for sending or manipulating messages on the bus

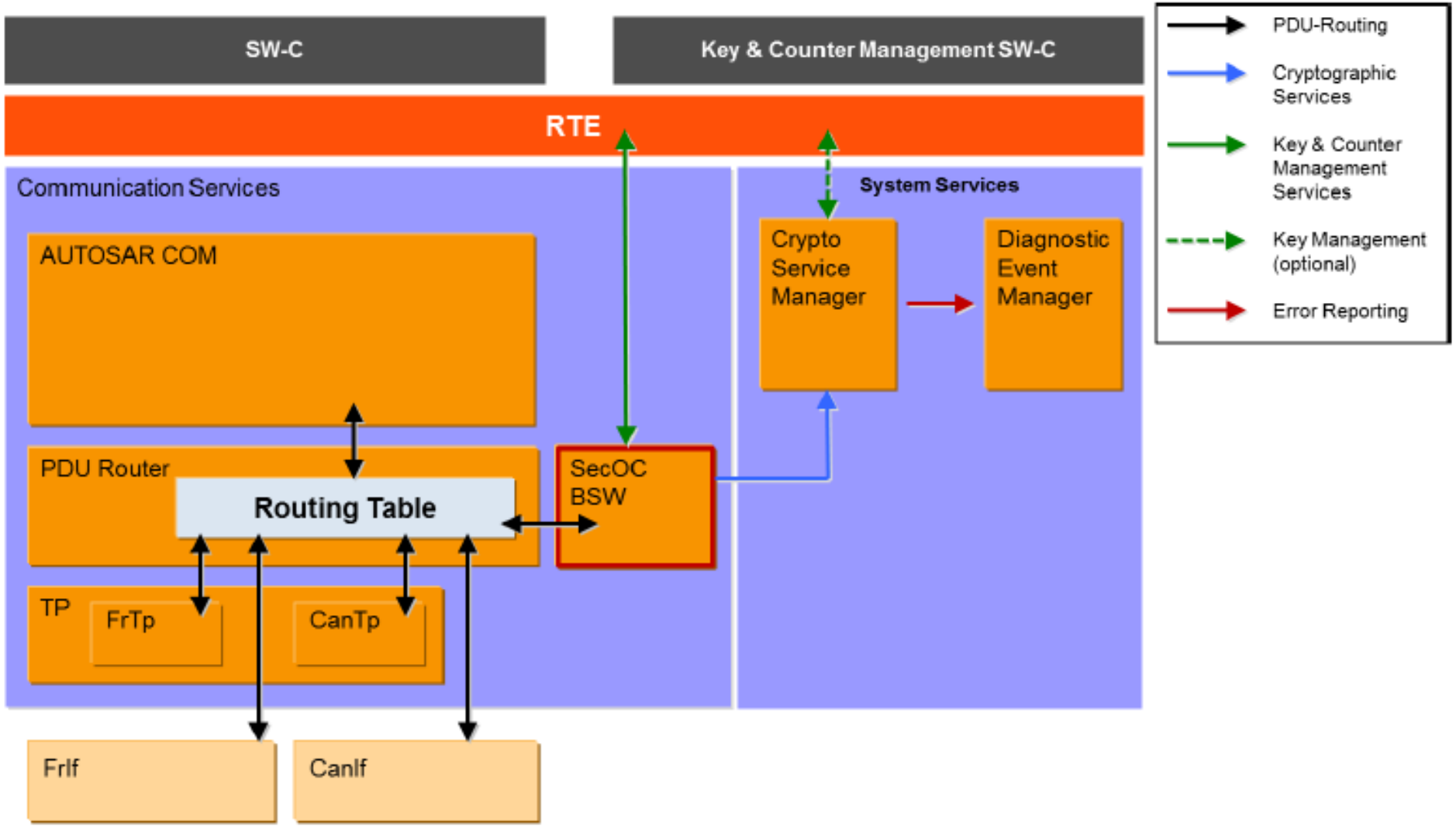


AUTOSAR Secure Onboard Communication

- AUTOSAR Secure Onboard Communication (SecOC) was specified to check the authenticity of a single transmitted protocol data unit, in order to detect attacks such as replay, spoofing and tampering.

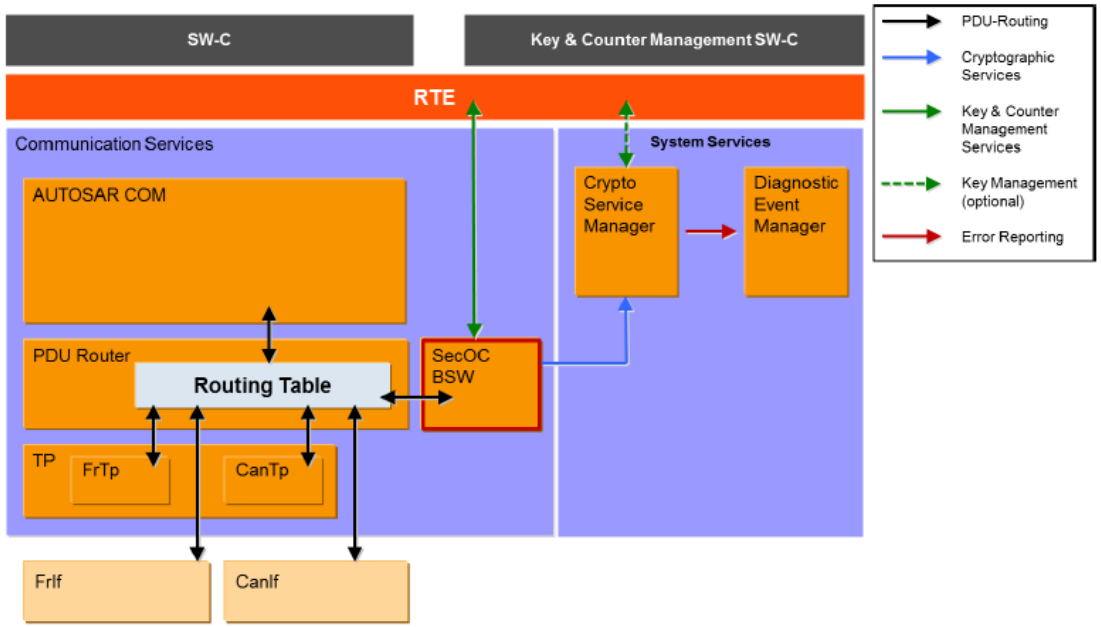
- AUTOSAR SecOC module provides authentication mechanism
  - A module located in the communication services area
  - Supports mainly symmetric authentication approaches with Message Authentication Codes (MACs) → efficiency and resource consumption
  - However, asymmetric authentication approaches can be used such as digital signature
  - Hardware/Software based implementation

AUTOSAR Secure Onboard Communication



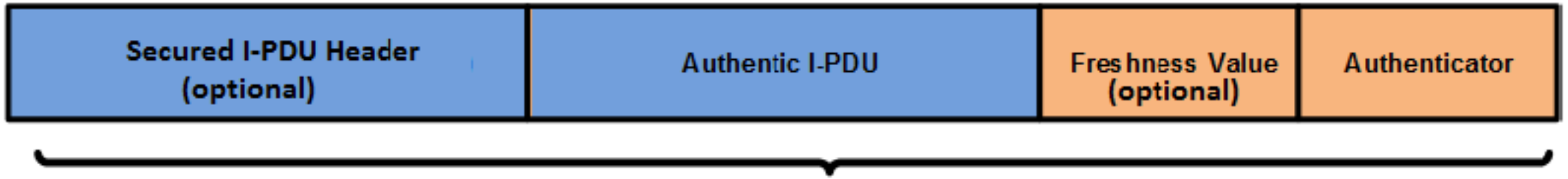
AUTOSAR Secure Onboard Communication

- SecOC interacts directly with the PduR, CSM, and RTE
- PDU Router (PduR): Route in/out PDU to the SecOC module
- The SecOC module
  - uses crypto services from the Crypto Service Manager (CSM) module
  - interacts with RTE module to allow key and counter management
  - add/process security relevant info and sends the PDUs to the PduR
  - supports all kind of communication paradigms supported by PduR



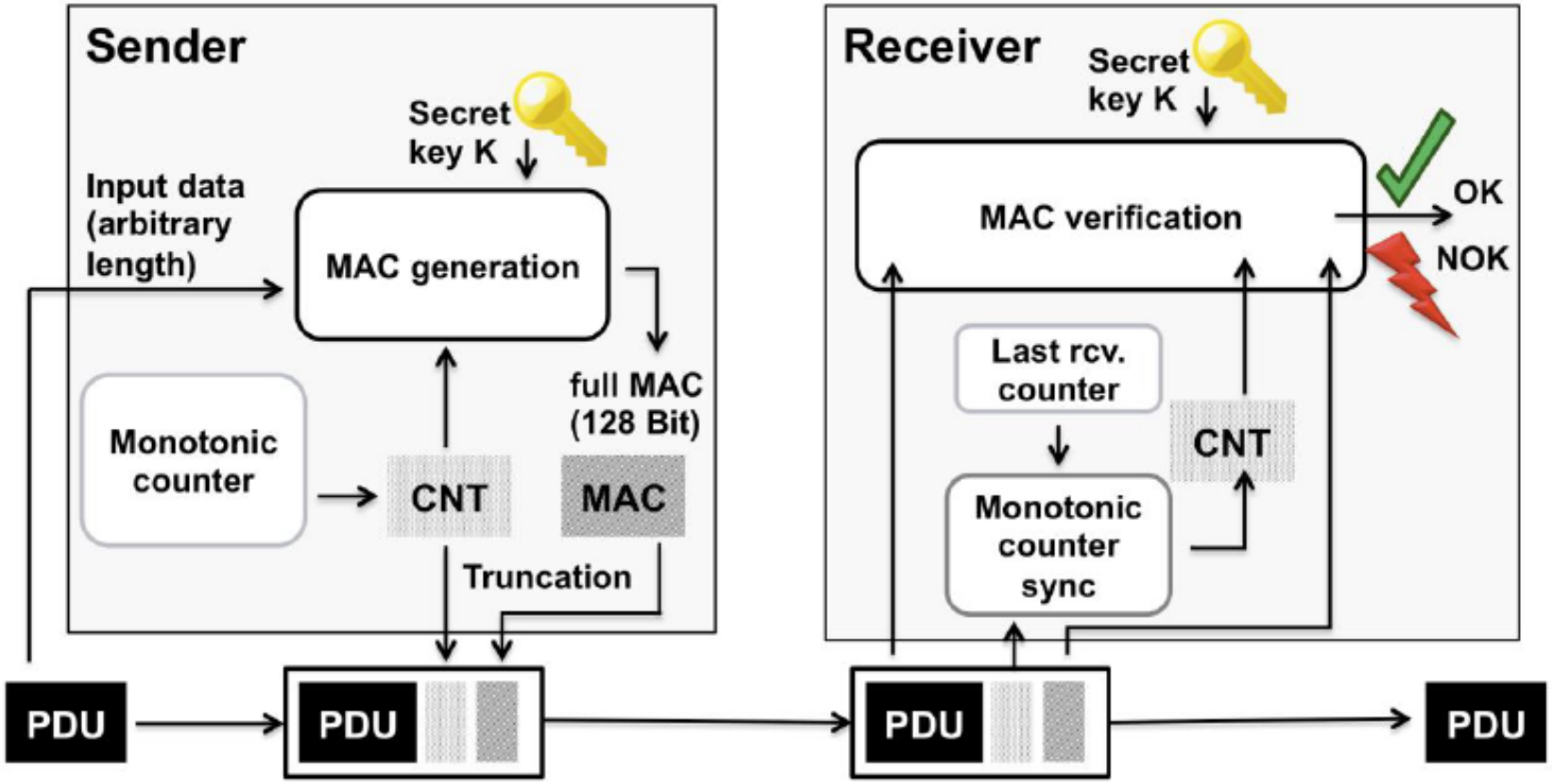
AUTOSAR SecOC: Terminologies

Terms	Definitions
I-PDU	An Interaction Layer Protocol Data Unit (I-PDU) is used for the data exchange of the modules directly above the PDU Router, for example, AUTOSAR COM
Authentic I-PDU	An AUTOSAR I-PDU that requires protection against unauthorized manipulation and replay attacks
Secured I-PDU	The payload of a Secured I-PDU consists of the Authentic I-PDU and an Authenticator (e.g. Message Authentication Code). The payload of a Secured I-PDU may optionally include the Freshness Value used to create the Authenticator (e.g. MAC).



Secured I-PDU Contents

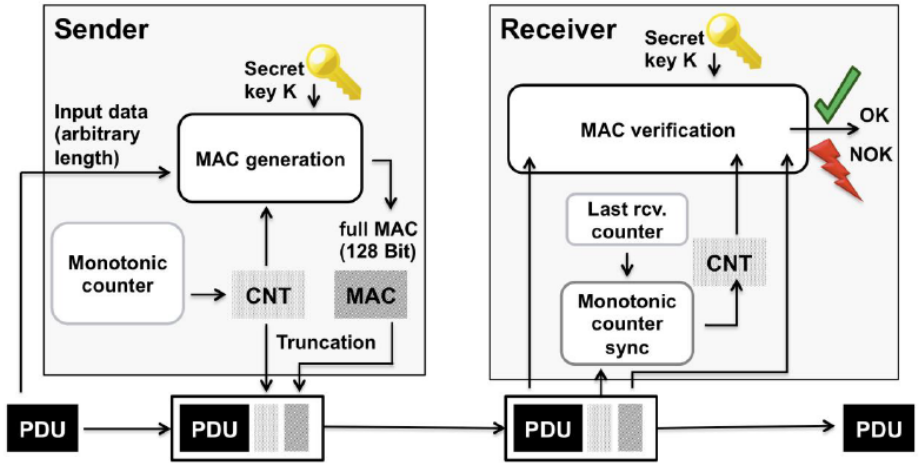
AUTOSAR SecOC: Security Solution



Message Authentication and Freshness Verification

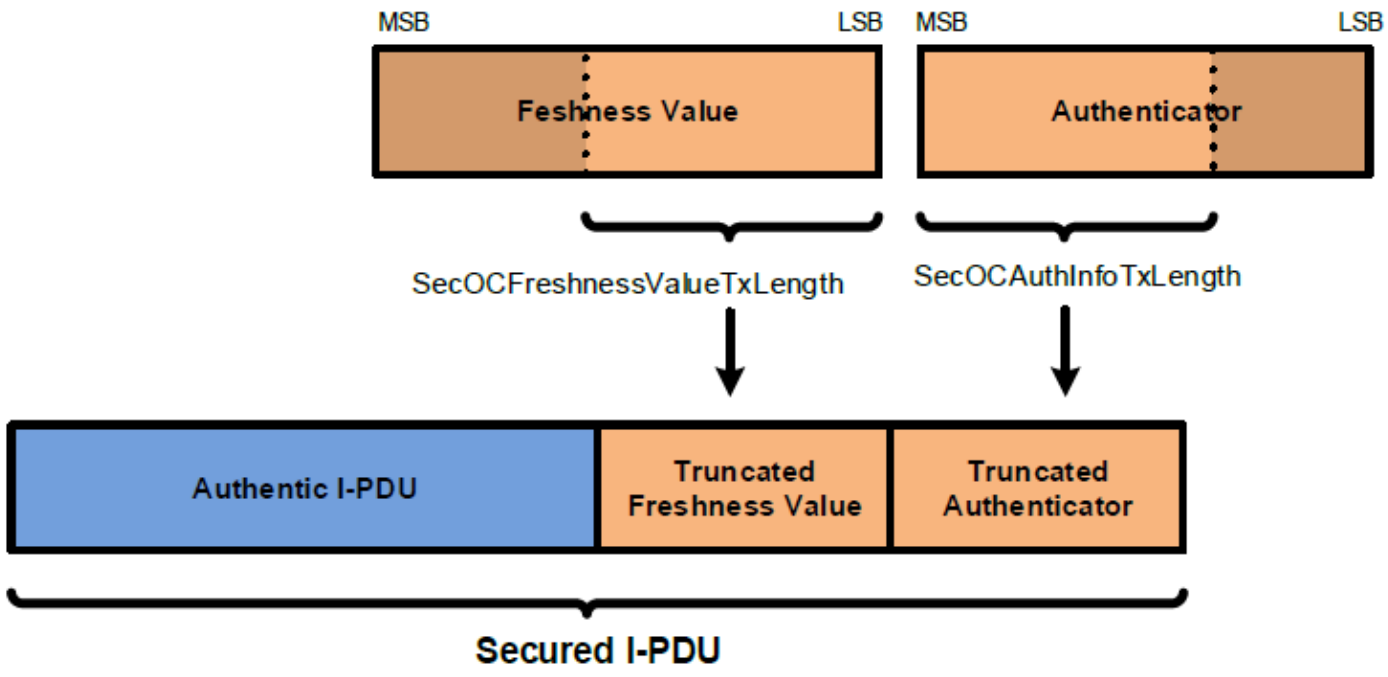
AUTOSAR SecOC: Security Solution

- Sender
  - Receives Freshness Value (FV) from the Freshness Manager (FM)
  - FM uses mechanisms: Timestamp or Freshness Counter
  - FV is considered during the generation of the Authenticator
  - Creates a Secured I-PDU by adding authentication information (Mandatory: Authenticator such as MAC, Optional: FV)
- Receiver
  - Receives FV from the FM
  - Verifies the authentication information added by the sender



AUTOSAR SecOC: Truncation

- Authentication mechanism may truncate the FV and the Authenticator due to limited payload length
- The Authenticator should only be truncated down to the most significant bits



AUTOSAR SecOC: Recommendations

- For the resource constraint embedded use case with static participants, AUTOSAR proposes using Message Authentication Codes (MACs) as a basis for authentication (e.g. a CMAC based on AES with an adequate key length
- In case a MAC is used, it is possible to transmit and compare only parts of the MAC. This is known as MAC truncation. However, this results in a lower security level at least for forgery of single MACs
- While AUTOSAR proposes to always use a key length of at least 128 bit, a MAC truncation can be beneficial. Of course, the actual length of the MAC for each use case has to be chosen carefully
- In general, MAC sizes of 64 bit and above are considered to provide sufficient protection against guessing attacks by NIST